

TUNICA, MS, USA

WMO: 722364

Lat: 34.681N

Lon: 90.347W

Elev: 194

StdP: 14.59

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 23903

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.3	26.6	9.7	9.1	25.8	14.1	11.3	29.3	26.5	47.6	24.7	47.4	8.1	0	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	96.8	75.4	94.7	75.8	91.2	74.9	79.6	89.9	78.4	88.8	77.6	88.1	6.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	142.2	84.5	75.4	133.8	83.8	73.4	125.3	82.0	43.3	87.9	42.0	88.5	40.9	87.8	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	19.9	17.8	17.4	98.9	3.9	2.9	14.6	101.0	12.4	102.7	10.2	104.3	7.3	106.4
			17.0	81.2	5.0	1.2	13.3	82.1	10.4	82.7	7.6	83.4	3.9	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	41.8	44.6	53.2	62.1	70.9	79.0	80.4	80.1	74.4	62.5	50.7	43.9
	DBStd	16.22	9.93	9.90	9.90	8.06	7.00	4.18	4.01	4.46	5.96	8.52	8.99	8.80
	HDD50	933	291	209	80	10	0	0	0	0	7	100	236	
	HDD65	3100	719	574	382	145	33	0	0	6	153	435	653	
	CDD50	5335	37	57	180	373	648	871	942	933	732	393	121	48
	CDD65	2022	0	1	16	58	215	421	477	468	287	74	5	0
	CDH74	20218	0	3	100	463	1850	4428	5022	4812	2745	749	45	1
	CDH80	8847	0	0	8	81	646	2046	2337	2310	1183	231	5	0
Wind	WSAvg	7.2	9.0	9.0	9.2	9.0	7.3	6.1	5.1	4.7	5.5	6.5	7.5	8.1
Precipitation	PrecAvg	52.0	5.2	4.3	5.2	5.3	5.1	3.9	4.1	3.0	3.0	3.7	4.7	5.4
	PrecMax	70.1	8.5	9.1	9.8	18.1	10.9	9.0	9.9	6.8	9.8	10.5	12.0	11.6
	PrecMin	35.3	2.5	1.8	1.2	1.4	0.8	0.6	0.8	0.2	0.2	0.0	1.4	1.2
	PrecStd	10.6	1.6	2.1	2.2	3.5	2.6	2.2	2.4	1.7	2.1	2.4	2.8	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.2	72.8	81.2	86.2	91.2	97.2	97.4	99.5	96.9	90.0	79.3	72.2
		MCWB	64.0	64.5	65.0	67.5	71.2	74.5	75.8	76.7	71.8	70.4	64.5	65.9
	2%	DB	65.5	69.8	77.1	81.8	89.8	94.8	95.3	97.0	92.6	86.2	73.1	67.6
		MCWB	60.9	63.3	64.3	66.4	71.3	74.7	77.0	76.1	73.4	69.9	62.4	62.4
	5%	DB	62.9	64.5	72.6	79.3	86.5	91.3	93.0	94.6	90.3	81.8	70.4	63.1
		MCWB	58.5	58.9	62.1	65.0	69.9	74.0	76.7	76.5	73.0	67.0	61.1	59.3
	10%	DB	58.8	61.4	69.7	76.7	83.6	90.3	90.6	90.7	87.8	78.6	65.8	59.1
		MCWB	53.6	55.7	59.9	64.0	69.9	73.7	76.3	75.9	72.4	64.7	59.1	54.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.1	66.5	68.0	71.8	75.4	79.2	81.1	81.3	77.5	74.7	68.5	66.4
		MCDB	69.2	70.9	76.9	80.5	85.1	89.7	90.5	92.5	88.6	84.9	75.0	71.3
	2%	WB	62.2	63.8	66.1	69.6	74.1	77.8	79.6	79.9	76.5	72.8	66.0	63.2
		MCDB	64.8	67.2	74.5	77.8	83.5	89.1	89.5	90.7	87.4	82.0	72.2	65.4
	5%	WB	59.3	60.8	63.8	67.6	72.8	76.8	78.5	78.5	75.4	70.7	63.2	60.0
		MCDB	62.5	64.8	70.4	76.0	81.9	87.9	88.6	88.9	85.8	77.7	67.0	62.8
	10%	WB	54.2	56.9	61.7	66.0	71.5	75.7	77.8	77.5	74.1	68.2	60.1	55.9
		MCDB	57.2	61.6	67.9	73.9	80.4	86.1	88.1	87.6	83.1	74.7	65.4	59.5
Mean Daily Temperature Range	5% DB	MDBR	15.9	16.4	19.1	20.1	19.4	19.9	18.7	19.4	21.9	22.5	20.2	15.2
		MCDBR	20.6	21.6	22.3	22.6	22.6	21.9	21.4	22.7	25.0	26.4	25.1	18.8
		MCWBR	17.0	15.3	13.1	10.6	8.4	7.2	7.1	6.8	9.0	11.8	15.4	15.4
	5% WB	MCDBR	19.1	19.1	19.7	18.9	18.7	19.5	18.8	19.4	20.5	21.1	20.4	18.0
MCWBR		17.4	15.3	12.8	10.9	8.6	7.7	7.6	7.7	8.3	11.5	15.2	15.8	
Clear-Sky Solar Irradiance	taub	0.320	0.329	0.365	0.404	0.441	0.466	0.499	0.491	0.424	0.363	0.337	0.319	
	taud	2.495	2.464	2.390	2.274	2.230	2.170	2.113	2.122	2.301	2.450	2.475	2.513	
	Ebn at Noon	282	291	288	280	270	262	252	252	265	275	272	275	
	Edh at Noon	27	31	36	42	45	48	50	49	39	31	27	25	
All-Sky Solar Radiation	RadAvg	782	946	1321	1672	1882	2075	1979	1833	1571	1235	883	677	
	RadStd	66	133	124	108	106	121	102	127	113	151	93	68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page